

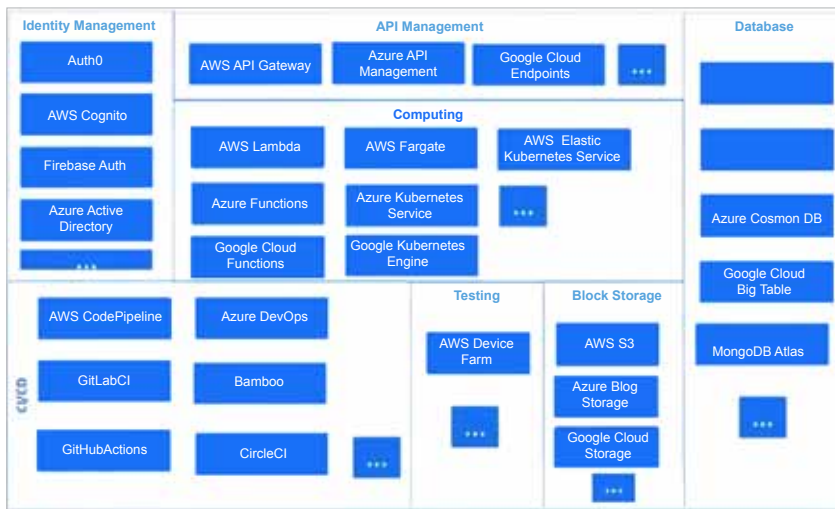


Figure 1: Serverless landscape

The advantages of serverless computing:

1. Event driven computing
2. Cost efficiency
3. Smaller DevOps cycle
4. Lower management and compliance burden
5. Faster iterations and rollout of software and services
6. Enhanced reliability
7. Enhanced security; updates to security are rolled out in a timely way and do not affect the availability of services
8. Eliminates the need to time the scaling of capacity and resources; sudden peaks and drops in traffic are managed efficiently

other leading industries come up with newer serverless based mechanisms. Its offerings are not just limited to computing but also affect databases, storage, identity management, API management, continuous integration and deployment (CI/CD), and so on.

Many startups, where cost plays a significant factor, have considered serverless as their primary choice for cloud services.

Serverless computing is expanding in all directions and is not just limited to one area (Figure 1). This landscape is predicted to only increase and not decrease, as serverless computing reduces costs and is very efficient.

Serverless catalogue

Identity management: From a client

perspective, identity management is crucial for any application and involves a lot of effort with respect to compliance. Serverless computing simplifies many things like saving development and maintenance time by integrating with infrastructure resources, both on the cloud as well as on-premise. Services available for this include AWS Cognito, Auth0, Microsoft Azure Active Directory, and Google Firebase Auth, which are especially powerful for sensitive data and help save a lot of time.

API management: With most enterprises migrating to an API-first approach for their solutions, API management is essential for building a distributed and scalable architecture. It provides a proxy for your computer

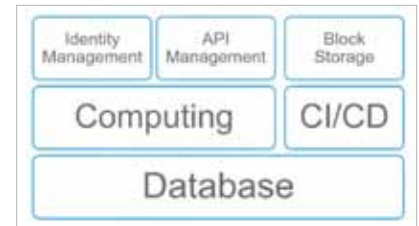


Figure 2: Serverless catalogue

storage layer. Many solutions also support microservices architecture. REST and GraphQL are popular for designing APIs. AWS API gateway, Azure API management, and Google Cloud Endpoints provide many default services. This is unlike yesteryears where developers had to take care of even minute services.

Computing: This involves two aspects, namely, serverless functions and containerisation. The functions expand and provide the ability to focus just on business logic and not on setting up the runtime environment. It works on event triggers like API invocation, message pushing in a queue, identity events, etc; integrates well with mobile as well as Web front-ends, and can scale up to millions of requests. Popular application runtimes like AWS Lambda, Azure Functions and Google Cloud Functions are supported by cloud providers.

Serverless containerisation involves containers that are fast becoming viable alternatives to virtual machines and reduce deployment issues. Cloud providers also provide serverless containerisation — for example, AWS Fargate, AWS Elastic Kubernetes Service, Azure Kubernetes Service and Google Kubernetes Engine. These services are quite powerful, and you only pay for what you use.

Database: Serverless computing offers the feasibility of databases, which are the fundamental building blocks of any application and help in storing structured and unstructured data. Both SQL and NoSQL databases are offered by cloud providers. They